



Chilli Black Thrips: The Tiny Pest with a Big Impact

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Abstract

Chilli black thrips (*Thrips parvispinus*) has recently emerged as a devastating pest in chilli growing regions of India, causing severe economic losses. The insect damages leaves, flowers, and young fruits, leading to leaf curling, flower drop, fruit scarring and reduced yields. Its rapid multiplication, hidden feeding habit, and resistance to insecticides make management highly challenging. Integrated strategies including field sanitation, crop rotation, use of biological control agents and judicious application of recommended insecticides such as spinosad, emamectin benzoate, and fipronil are essential for effective suppression. Adoption of Integrated Pest Management (IPM) can provide long-term and sustainable solutions to farmers facing this pest menace.

Keywords: Chilli, Black thrips, Pest management, IPM and Insecticides

Introductuon : Chilli, often called the “queen of spices,” is one of the most important cash crops in India and many other countries. It is not only consumed domestically but also exported widely, making it a crucial income source for farmers. In recent years, however, chilli farmers have faced a severe challenge from a minute insect pest the chilli black thrips (*Thrips parvispinus*). Despite being less than 2 mm in size, this pest has devastated chilli fields, particularly in Andhra Pradesh, Telangana and Karnataka (Kumar *et al.*, 2022).

Meet the Pest: *Thrips parvispinus*

The chilli black thrips is a small, slender insect that feeds on tender plant parts by sucking cell sap. It was once a minor pest but has emerged as a major threat due to favourable climatic conditions and resistance to commonly used insecticides (Reddy *et al.*, 2021). The pest multiplies rapidly, completing its life cycle in just 15-20 days, making it difficult to manage once established.

Damage Symptoms

Typical signs of infestation include:

- Upward leaf curling and distortion
- Silvery patches on leaves due to cell sap loss
- Flower drying and dropping
- Corky scarring on young fruits
- Stunted plant growth and reduced yields

These symptoms are often confused with nutrient deficiencies or viral diseases, delaying timely control.

Why It Is a Major Threat

- Thrives under warm, dry climates common in chilli-growing areas
- Rapid multiplication within 2-3 weeks
- Resistance to insecticides due to repeated spraying
- Hidden feeding habit inside curled leaves and buds
- Can transmit viruses, worsening crop loss (Kakkar *et al.*, 2019)



Fig 1: Damage symptoms

Management Strategies

1. Cultural Practices

- Crop rotation with non-host crops
- Removal and destruction of infested crop residues
- Installation of yellow and blue sticky traps (12-15 traps/acre) to monitor thrips
- Maintaining field sanitation

2. Biological Control

- Predatory mites (*Amblyseius swirskii*) and minute pirate bugs (*Orius* spp.) feed on thrips larvae.
- Neem-based products (azadirachtin 0.03% at 2 ml/L) reduce thrips populations.

3. Chemical Management (Recommended Sprays)

- Spinosad 45 SC @ 0.3 ml/L of water
- Fipronil 5 SC @ 2 ml/L
- Emamectin benzoate 5 SG @ 0.4 g/L
- Lambda-cyhalothrin 5 EC @ 0.5 ml/L
- Thiamethoxam 25 WG @ 0.25 g/L

Rotate chemicals from different groups to avoid resistance build-up. Ensure thorough coverage, especially on the undersides of leaves and flower buds.

4. Integrated Pest Management (IPM)

A combination of cultural, biological, and chemical methods is essential. Farmer awareness and training in safe pesticide handling are equally important for successful IPM adoption.

Conclusion

Chilli black thrips has become a serious menace to chilli farmers, threatening both yield and quality. By adopting integrated management practices that include sanitation, biological agents, and careful use of recommended insecticides, farmers can bring thrips populations under control. Long-term solutions must focus on sustainable IPM to ensure chilli remains a profitable crop for future generations.

References

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